

TucTuc Analysis

Luís Felipe Romera

TucTuc Analysis

In this document I'm going to adjust the data of **TucTuc Nanostructured Lipid Carrier** into a **multiple regression**, known as *scheffé model*.

```
library(magick)
```

Warning: pacote 'magick' foi compilado no R versão 4.5.2

Linking to ImageMagick 6.9.13.29

Enabled features: cairo, freetype, fftw, ghostscript, heic, lcms, pango, raw, rsvg, webp

Disabled features: fontconfig, x11

```
getwd()
```

```
[1] "C:/Users/luisr/Documentos/Documents/Documentos - Copia/Projeto_TucTuc/TucumaButterOil_ar"
```

```
data <- read.csv("NanoTucumaTucuma.csv")
```

```
data
```

	Tucuma	Tucuma.1	Brijo10	PDI	Size
1	0.2500	0.2500	0.5000	0.104	31.00
2	0.1665	0.5000	0.3335	0.131	92.00
3	0.5000	0.1665	0.3335	0.135	88.00
4	0.3750	0.3750	0.2500	0.252	105.00
5	0.2500	0.5000	0.2500	0.299	150.00
6	0.2500	0.1665	0.5835	0.071	22.09
7	0.5000	0.2500	0.2500	0.384	138.00

8	0.3335	0.1665	0.5000	0.081	28.00
9	0.1650	0.2500	0.5835	0.177	24.14
10	0.3335	0.5000	0.1650	0.581	382.00
11	0.3335	0.3335	0.3335	0.135	96.00
12	0.3335	0.3335	0.3335	0.117	84.20
13	0.3335	0.3335	0.3335	0.280	73.77
14	0.3335	0.3335	0.3335	0.118	70.60
15	0.3335	0.3335	0.3335	0.138	89.64

Regression models

First, let's fit a *linear Scheffé model*, given by:

$$\sum_{i=1}^q \beta_i x_i + \epsilon$$

```
library(mixexp)
?MixModel

linMod <- MixModel(data,
  response = "PDI",
  mixcomps = c("Tucuma", "Tucuma.1", "Brijo10"),
  model = 1)
```

	coefficients	Std.err	t.value	Prob
Tucuma	0.4026537	0.2065288	1.949624	0.07497682
Tucuma.1	0.5348577	0.1785973	2.994770	0.01117478
Brijo10	-0.2749660	0.1534255	-1.792179	0.09833306

Residual standard error: 0.1084305 on 12 degrees of freedom
 Corrected Multiple R-squared: 0.4757822

```
summary(linMod)
```

Call:

```
lm(formula = mixmodnI, data = frame)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.112449	-0.084371	-0.004709	0.050324	0.224686

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
Tucuma	0.3996	0.2067	1.933	0.0771 .
Tucuma.1	0.5364	0.1789	2.999	0.0111 *
Brijo10	-0.2738	0.1537	-1.781	0.1001

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1086 on 12 degrees of freedom

Multiple R-squared: 0.8372, Adjusted R-squared: 0.7966

F-statistic: 20.58 on 3 and 12 DF, p-value: 5.055e-05

```
anova(linMod)
```

Analysis of Variance Table

Response: PDI

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Tucuma	1	0.61039	0.61039	51.7103	1.1e-05 ***
Tucuma.1	1	0.08084	0.08084	6.8490	0.02251 *
Brijo10	1	0.03746	0.03746	3.1735	0.10015
Residuals	12	0.14165	0.01180		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

It is possible to see that only the coefficient for **Tucumã butter** have a significant *p-value* for the **t** estimate, meaning that it's the only coefficient that is significant different from zero, contributing individually to the model.

In the *ANOVA table*, **Brij O10** does not have a significant *p-value* for the **F** estimate, meaning that a model without it would still explain the same variance.

Also, we can see that the *corrected Multiple R-squared* is only 0.48 approximately, indicating **lack of fit**.

Now, let's fit a *quadratic* model, given by:

$$\sum_{i=1}^q \beta_i x_i \sum_{i=1}^{q-1} \sum_{j=i+1}^q \beta_{ij} x_i x_j + \epsilon$$

```
quadMod <- MixModel(data,
  response = "PDI",
  mixcomps = c("Tucuma", "Tucuma.1", "Brijo10"),
  model = 2)
```

	coefficients	Std.err	t.value	Prob
Tucuma	1.6359042	0.6127583	2.6697380	0.025633457
Tucuma.1	0.9159281	0.5456683	1.6785435	0.127549373
Brijo10	1.7643905	0.3917292	4.5041082	0.001480184
Tucuma.1:Tucuma	0.2003188	1.6778403	0.1193909	0.907588244
Brijo10:Tucuma	-6.8718548	1.8376561	-3.7394672	0.004629298
Tucuma.1:Brijo10	-4.8633271	1.4662879	-3.3167614	0.008986562

Residual standard error: 0.06092945 on 9 degrees of freedom
Corrected Multiple R-squared: 0.8758563

```
summary(quadMod)
```

Call:

```
lm(formula = mixmodnI, data = frame)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.07418	-0.03090	-0.01189	0.03201	0.12419

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
Tucuma	1.6481	0.6211	2.654	0.02632 *
Tucuma.1	0.9253	0.5476	1.690	0.12538
Brijo10	1.7845	0.3966	4.500	0.00149 **
Tucuma:Tucuma.1	0.1865	1.6882	0.110	0.91447
Tucuma:Brijo10	-6.9378	1.8685	-3.713	0.00482 **
Tucuma.1:Brijo10	-4.9150	1.4706	-3.342	0.00863 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.06109 on 9 degrees of freedom
Multiple R-squared: 0.9614, Adjusted R-squared: 0.9357
F-statistic: 37.36 on 6 and 9 DF, p-value: 7.308e-06

```
anova(quadMod)
```

Analysis of Variance Table

Response: PDI

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Tucuma	1	0.61039	0.61039	163.5409	4.47e-07 ***
Tucuma.1	1	0.08084	0.08084	21.6608	0.001195 **
Brijo10	1	0.03746	0.03746	10.0366	0.011402 *
Tucuma:Tucuma.1	1	0.02016	0.02016	5.4023	0.045168 *
Tucuma:Brijo10	1	0.04620	0.04620	12.3793	0.006532 **
Tucuma.1:Brijo10	1	0.04169	0.04169	11.1701	0.008630 **
Residuals	9	0.03359	0.00373		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The *quadratic model* seems to fit the data better than the *linear model*. The coefficient for **Tucumã butter** and the coefficient for the interaction of **Tucumã butter** and **Tucumã oil** are the only ones that don't have a significant *p-value* for the *t* estimate. The *Corrected Multiple R-squared* is of approximately 0.88, which indicates good fit and all the coefficients have a significant *p-value* for the **F** estimate.

Now let's try fitting a *scheffé's special cubic model*, given by:

$$\sum_{i=1}^q \beta_i x_i + \sum_{i=1}^{q-1} \sum_{j=i+1}^q \beta_{ij} x_i x_j + \sum_{i=1}^{q-2} \sum_{j=i+1}^{q-1} \sum_{k=j+1}^q \beta_{ijk} x_i x_j x_k + \epsilon$$

to see if it explains the variance even better than the *quadratic model*.

```
cubMod <- MixModel(data,
  response = "PDI",
  mixcomps = c("Tucuma", "Tucuma.1", "Brijo10"),
  model = 4)
```

	coefficients	Std.err	t.value	Prob
Tucuma	0.7695780	2.102651	0.36600362	0.7238519
Tucuma.1	-0.1826307	2.602456	-0.07017628	0.9457759
Brijo10	0.9975264	1.819208	0.54833018	0.5984290
Tucuma.1:Tucuma	5.0160362	11.267397	0.44518144	0.6679882
Brijo10:Tucuma	-3.0554274	9.027838	-0.33844506	0.7437390
Tucuma.1:Brijo10	-0.3013074	10.654420	-0.02828004	0.9781316
Tucuma.1:Brijo10:Tucuma	-15.0430514	34.764808	-0.43270918	0.6766545

Residual standard error: 0.0638822 on 8 degrees of freedom
Corrected Multiple R-squared: 0.8786954

```
summary(cubMod)
```

Call:
lm(formula = mixmodnI, data = frame)

Residuals:

Min	1Q	Median	3Q	Max
-0.07968	-0.03106	-0.01660	0.02512	0.12540

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
Tucuma	0.72538	2.07514	0.350	0.736
Tucuma.1	-0.24057	2.55499	-0.094	0.927
Brijo10	0.96291	1.80306	0.534	0.608
Tucuma:Tucuma.1	5.28286	11.02681	0.479	0.645
Tucuma:Brijo10	-2.86734	8.91042	-0.322	0.756
Tucuma.1:Brijo10	-0.06668	10.46821	-0.006	0.995
Tucuma:Tucuma.1:Brijo10	-15.89158	33.93971	-0.468	0.652

Residual standard error: 0.06393 on 8 degrees of freedom
Multiple R-squared: 0.9624, Adjusted R-squared: 0.9296
F-statistic: 29.28 on 7 and 8 DF, p-value: 4.329e-05

```
anova(cubMod)
```

Analysis of Variance Table

Response: PDI

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Tucuma	1	0.61039	0.61039	149.3536	1.865e-06 ***
Tucuma.1	1	0.08084	0.08084	19.7817	0.002146 **
Brijo10	1	0.03746	0.03746	9.1659	0.016371 *
Tucuma:Tucuma.1	1	0.02016	0.02016	4.9337	0.057077 .
Tucuma:Brijo10	1	0.04620	0.04620	11.3054	0.009897 **
Tucuma.1:Brijo10	1	0.04169	0.04169	10.2010	0.012728 *
Tucuma:Tucuma.1:Brijo10	1	0.00090	0.00090	0.2192	0.652113
Residuals	8	0.03269	0.00409		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
anova(quadMod, cubMod)
```

Analysis of Variance Table

Model 1: PDI ~ -1 + Tucuma + Tucuma.1 + Brijo10 + Tucuma:Tucuma.1 + Tucuma:Brijo10 + Tucuma.1:Brijo10

Model 2: PDI ~ -1 + Tucuma + Tucuma.1 + Brijo10 + Tucuma:Tucuma.1 + Tucuma:Brijo10 + Tucuma.1:Brijo10 + Tucuma:Tucuma.1:Brijo10

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	9	0.033591				
2	8	0.032695	1	0.000896	0.2192	0.6521

In the *special cubic model*, none of the coefficients have a significant *p-value* for the **t** estimate, and some doesn't have significant *p-values* for the **F** estimate, although the *Corrected Multiple R-squared* is greater than the quadratic model, indicating **overfitting**. Nonetheless, the *anova* performed showed that the *special cubic model* does not explain the variance significantly better than the *quadratic model*, indicating that the *quadratic model* is the best fit for the data.

Hence, substituting the β for the estimated coefficients, the model presents itself as:

$$PDI = 1.64x_1 + 0.92x_2 + 1.76x_3 + 0.2x_1x_2 - 6.87x_1x_3 - 4.86x_2x_3$$

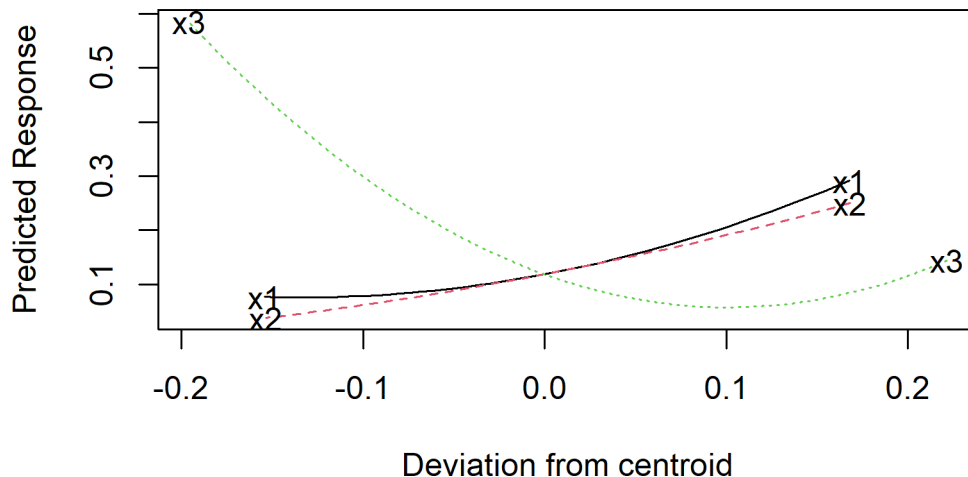
x_1 being **Tucumã oil**, x_2 being **Tucumã butter** and x_3 being **Brij O10**.

Here, we're able to see that the coefficients β_{13} and β_{23} are the only ones that contribute to a lower PDI. Also, they are the coefficients that have the greatest effect on the model. Therefore, the interaction between the Tucumã butter and the Tucumã oil with the surfactant Brij O10

has great influence in bringing the PDI value down. But the coefficient by itself does not indicate the response, because it is also dependent in which direction the other components will variate. For that matter, plotting the *Piepel effects directions* is fundamental to understand what happens to the response once we dislocate the formulation towards a component.

```
PDIEff <- ModelEff(nfac = 3,
  mod = 2,
  dir = 1,
  ufunc = quadMod,
  dimensions = list(x1="Tucuma", x2="Tucuma.1", x3="Brijo10"),
  lc = c(0.165, 0.1665, 0.165),
  uc = c(0.5, 0.5, 0.5835)
)
```

Effect Plot (Piepel direction)



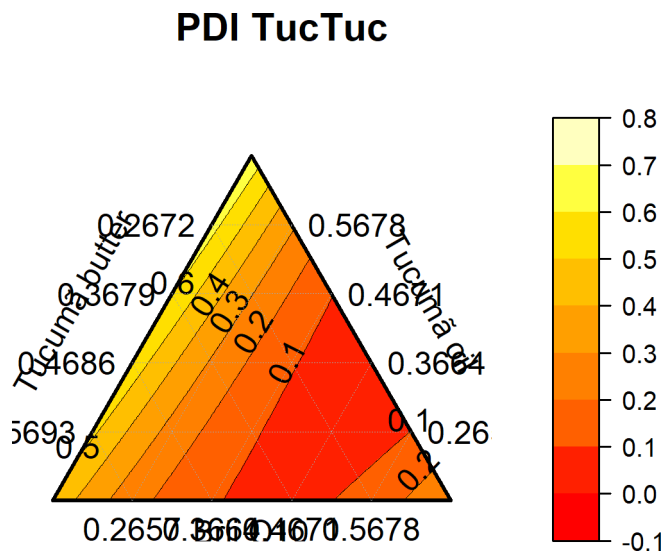
The Piepel Effect direction plot indicates that shifting the formulation in the direction of the surfactant and away from the other components tends to form more homogenic particles, but only till the parabole formed by the surfactant effect line reaches its minimum point, after that, the particles starts a heterogenic tendency again. Plotting a ternary graph of the PDI will help to visualize in which proportions of the components the nanoparticles tend to be more homogenic.

Now, let's plot a ternary graph to access how the PDI is molded by the components of the nanoparticle.

```

PDIPlot <- ModelPlot(model = quadMod,
  dimensions = list(x1="Tucuma", x2="Tucuma.1", x3="Brijo10"),
  lims = c(0.165, 0.5, 0.1665, 0.5, 0.165, 0.5835),
  contour = TRUE,
  pseudo = TRUE,
  fill = TRUE,
  main = "PDI TucTuc",
  axislabs = c("Tucumã oil", "Tucumã butter", "Brij O10"),
  axislab.offset = 0.1,
  cornerlabs = c("", "", ""),
  grid = TRUE,
  colorkey = TRUE,
  labels = TRUE,
  label.style = "mixed"
)

```



Analyzing the ternary graph, and according to the model fitted, we're able to see a region where the components proportions tend to generate lower PDI values. Hence, the optimal formulation, in seeking the lowest PDI, would have **27% Tucumã oil**, **27% Tucumã butter** and **46% Brij O10** in this part of the mixture.

Size

Now, let's do the same analysis for the size response.

This time I'll do the analysis quicker since the explanation was given in the **PDI** part.

```
linModSize <- MixModel(data,
  response = "Size",
  mixcomps = c("Tucuma", "Tucuma.1", "Brijo10"),
  model = 1)
```

	coefficients	Std.err	t.value	Prob
Tucuma	200.4859	113.25895	1.770155	0.102079436
Tucuma.1	366.2098	97.94147	3.739068	0.002826767
Brijo10	-229.1268	84.13747	-2.723244	0.018496413

Residual standard error: 59.46252 on 12 degrees of freedom
Corrected Multiple R-squared: 0.6047647

```
summary(linModSize)
```

Call:

```
lm(formula = mixmodnI, data = frame)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-50.095	-33.547	-16.532	9.366	169.856

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
Tucuma	198.87	113.32	1.755	0.10474
Tucuma.1	367.04	98.07	3.743	0.00281 **
Brijo10	-228.48	84.26	-2.712	0.01889 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 59.57 on 12 degrees of freedom
Multiple R-squared: 0.8312, Adjusted R-squared: 0.789
F-statistic: 19.7 on 3 and 12 DF, p-value: 6.269e-05

```
anova(linModSize)
```

Analysis of Variance Table

Response: Size

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Tucuma	1	148848	148848	41.9482	3.038e-05 ***
Tucuma.1	1	34762	34762	9.7966	0.008693 **
Brijo10	1	26094	26094	7.3537	0.018893 *
Residuals	12	42580	3548		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
quadModSize <- MixModel(data,
  response = "Size",
  mixcomps = c("Tucuma", "Tucuma.1", "Brijo10"),
  model = 2)
```

	coefficients	Std.err	t.value	Prob
Tucuma	438.7778	348.4452	1.2592447	0.239619506
Tucuma.1	1043.9191	310.2945	3.3642855	0.008332248
Brijo10	747.9402	222.7569	3.3576520	0.008420513
Tucuma.1:Tucuma	-237.7363	954.1044	-0.2491723	0.808820979
Brijo10:Tucuma	-2042.3151	1044.9837	-1.9543990	0.082382542
Tucuma.1:Brijo10	-3742.0742	833.8050	-4.4879486	0.001514984

Residual standard error: 34.64755 on 9 degrees of freedom
Corrected Multiple R-squared: 0.8993589

```
summary(quadModSize)
```

Call:

```
lm(formula = mixmodnI, data = frame)
```

Residuals:

Min	1Q	Median	3Q	Max
-62.332	-6.843	4.868	14.738	50.541

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
Tucuma	437.1	353.1	1.238	0.24712
Tucuma.1	1045.8	311.4	3.359	0.00841 **
Brijo10	751.8	225.5	3.334	0.00874 **
Tucuma:Tucuma.1	-232.6	959.9	-0.242	0.81397
Tucuma:Brijo10	-2047.2	1062.4	-1.927	0.08609 .
Tucuma.1:Brijo10	-3755.6	836.1	-4.492	0.00151 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 34.74 on 9 degrees of freedom

Multiple R-squared: 0.957, Adjusted R-squared: 0.9283

F-statistic: 33.35 on 6 and 9 DF, p-value: 1.185e-05

```
anova(quadModSize)
```

Analysis of Variance Table

Response: Size

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Tucuma	1	148848	148848	123.3650	1.484e-06 ***
Tucuma.1	1	34762	34762	28.8107	0.0004518 ***
Brijo10	1	26094	26094	21.6265	0.0012014 **
Tucuma:Tucuma.1	1	4055	4055	3.3608	0.0999789 .
Tucuma:Brijo10	1	3324	3324	2.7552	0.1313071
Tucuma.1:Brijo10	1	24342	24342	20.1746	0.0015070 **
Residuals	9	10859	1207		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
cubModSize <- MixModel(data,  
  response = "Size",  
  mixcomps = c("Tucuma", "Tucuma.1", "Brijo10"),  
  model = 4)
```

	coefficients	Std.err	t.value	Prob
Tucuma	-1093.2785	1067.4661	-1.0241810	0.3357188
Tucuma.1	-898.8294	1321.2052	-0.6803102	0.5155096
Brijo10	-608.2222	923.5684	-0.6585568	0.5286742
Tucuma.1:Tucuma	8278.6289	5720.1893	1.4472649	0.1858459
Brijo10:Tucuma	4706.8537	4583.2186	1.0269756	0.3344791
Tucuma.1:Brijo10	4325.6389	5408.9954	0.7997121	0.4469709
Tucuma.1:Brijo10:Tucuma	-26602.9148	17649.2650	-1.5073101	0.1701588

Residual standard error: 32.43147 on 8 degrees of freedom

Corrected Multiple R-squared: 0.9216189

```
summary(cubModSize)
```

Call:

```
lm(formula = mixmodnI, data = frame)
```

Residuals:

Min	1Q	Median	3Q	Max
-71.516	-5.758	0.148	14.924	33.291

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
Tucuma	-1103.1	1049.5	-1.051	0.324
Tucuma.1	-900.1	1292.2	-0.697	0.506
Brijo10	-619.5	911.9	-0.679	0.516
Tucuma:Tucuma.1	8273.9	5576.8	1.484	0.176
Tucuma:Brijo10	4746.8	4506.5	1.053	0.323
Tucuma.1:Brijo10	4336.7	5294.3	0.819	0.436
Tucuma:Tucuma.1:Brijo10	-26524.8	17165.1	-1.545	0.161

Residual standard error: 32.33 on 8 degrees of freedom

Multiple R-squared: 0.9669, Adjusted R-squared: 0.9378

F-statistic: 33.33 on 7 and 8 DF, p-value: 2.649e-05

```
anova(cubModSize)
```

Analysis of Variance Table

Response: Size

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Tucuma	1	148848	148848	142.3889	2.238e-06	***
Tucuma.1	1	34762	34762	33.2535	0.0004209	***
Brijo10	1	26094	26094	24.9615	0.0010579	**
Tucuma:Tucuma.1	1	4055	4055	3.8791	0.0844049	.
Tucuma:Brijo10	1	3324	3324	3.1801	0.1123848	
Tucuma.1:Brijo10	1	24342	24342	23.2857	0.0013121	**
Tucuma:Tucuma.1:Brijo10	1	2496	2496	2.3879	0.1608619	
Residuals	8	8363	1045			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
anova(linModSize, quadModSize)
```

Analysis of Variance Table

Model 1: Size ~ -1 + Tucuma + Tucuma.1 + Brijo10

Model 2: Size ~ -1 + Tucuma + Tucuma.1 + Brijo10 + Tucuma:Tucuma.1 + Tucuma:Brijo10 +
Tucuma.1:Brijo10

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	12	42580				
2	9	10859	3	31721	8.7636	0.004914 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The analysis of the statistical parameters of the models indicates that the **quadratic model** presents the best fit for the data. Thus, the model becomes:

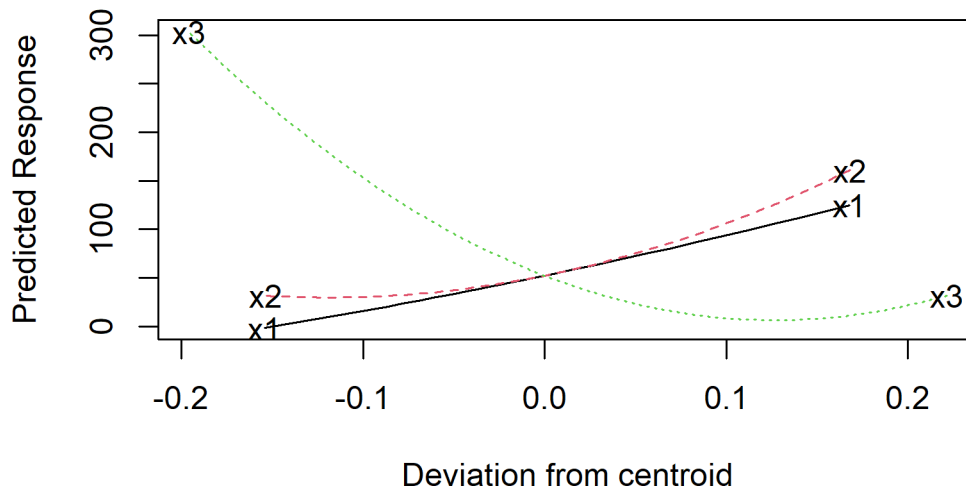
$$Size(nm) = 439x_1 + 1044x_2 + 748x_3 - 238x_1x_2 - 2042x_1x_3 - 3742x_2x_3$$

The coefficients demonstrate that the components by themselves tends to contribute to a higher size, whereas the interaction between the components tends to contribute to a lower size, with the interaction of the lipids with the surfactant having the greatest coefficient in module, indicating a strong effect towards reducing the size.

```
sizeEff <- ModelEff(nfac = 3,
  mod = 2,
  dir = 1,
  ufunc = quadModSize,
  dimensions = list(x1="Tucuma", x2="Tucuma.1", x3="Brijo10"),
  lc = c(0.165, 0.1665, 0.165),
```

```
uc = c(0.5, 0.5, 0.5835)
)
```

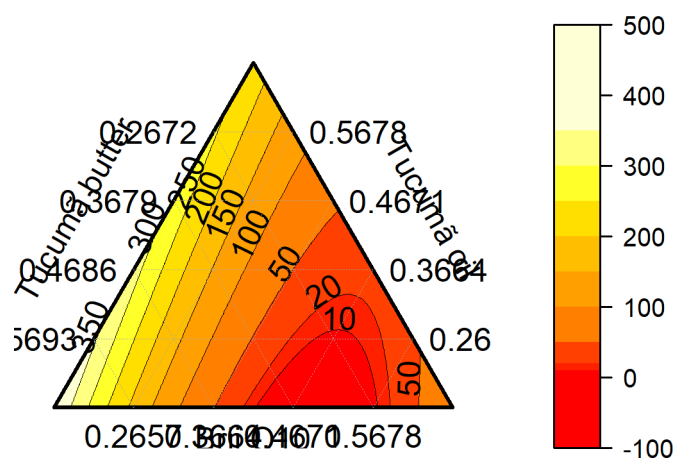
Effect Plot (Piepel direction)



Analyzing the Piepel effect directions, we're able to see a pattern similar to the PDI, where dislocating towards the surfactant and away from the lipids tends to generate a smaller size, till the inflection point on the parable described by the surfactant effect direction.

```
sizePlot <- ModelPlot(model = quadModSize,
  dimensions = list(x1="Tucuma", x2="Tucuma.1", x3="Brijo10"),
  lims = c(0.165, 0.5, 0.1665, 0.5, 0.165, 0.5835),
  contour = TRUE,
  at = c(-100, 10, 20, 50, 100, 150, 200, 250, 300, 350, 500),
  pseudo = TRUE,
  fill = TRUE,
  main = "PDI TucTuc",
  axislabs = c("Tucumã oil", "Tucumã butter", "Brij 010"),
  axislab.offset = 0.1,
  cornerlabs = c("", "", ""),
  grid = TRUE,
  colorkey = TRUE,
  labels = TRUE,
  label.style = "align"
)
```

PDI TucTuc



Analyzing the ternary graph, we can find an optimal region for a small size, according to the model, around **27% of Tucumã oil**, **27% of Tucumã butter** and **46% of Brij O10**, similar to the region of optimal PDI.